

Geometry and Fraction Puzzles

Mission

Warm-up and worked example

Name: _____

Date: _____

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YOUR MISSION Use spatial visualization, decomposition, symmetry, and fractions in non-routine puzzles.

SMART MOVE Draw a simpler case and organize what you know.

Math Talk

How many different ways can a square be divided into four equal areas?

My idea: _____

New Words

represent	deduce	assumption
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

- 1. Build a square from tangram pieces.
- 2. Identify fractional parts of the whole.

What the example shows: _____

Geometry and Fraction Puzzles

Mission

Learn it and show your thinking

Name: _____

Date: _____

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YOUR MISSION Follow the learning path, then explain the big idea.

SMART MOVE Draw before asking for a rule.

[] **Step 1:** Build a square from tangram pieces.

[] **Step 2:** Identify fractional parts of the whole.

[] **Step 3:** Rearrange pieces without changing total area.

[] **Step 4:** Solve a perimeter puzzle on grid paper.

[] **Step 5:** Discuss what changes and what remains invariant.

My model, drawing, or organized work:

The big idea in my own words:

Geometry and Fraction Puzzles

Mission

Practice lab

Name: _____

Date: _____

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YOUR MISSION Use a model, equation, or explanation for every task.

SMART MOVE Check whether your answer is reasonable.

1. Complete two tangram silhouettes.

2. Partition a rectangle into equal areas in three ways.

3. Find a missing area.

4. Solve one visual fraction comparison.

Choose one answer and prove it another way:

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Challenge and final check

Name: _____

Date: _____

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YOUR MISSION Try an unfamiliar problem and defend your reasoning.

SMART MOVE A wrong first idea can still lead to a strong solution.

Challenge Mission

Move one square tile in a shape so the perimeter changes but the area stays the same.

My plan:

My solution and proof:

Final Check

Name one quantity that stayed invariant during a puzzle.

☐ I can teach it

☐ I need practice

☐ I need help

Geometry and Fraction Puzzles

Mission

Reusable math tool

Name: _____

Date: _____

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YOUR MISSION Use this page to draw, model, organize, and check.

SMART MOVE Draw a simpler case and organize what you know.

Problem-Solving Organizer

What I know

What I need to find

My model or table

My equations

My check and explanation
